Introduction to Cybersecurity - Research Assignment

Assignment Title: Exploring the Domains of Cybersecurity: Roles, Skills, and Pathways

Due Date: 02/03/2026

Objective:

The primary objective of this assignment is to provide students with a comprehensive understanding of the diverse landscape of cybersecurity. By researching specific domains, students will explore various career paths, the technical and soft skills required, essential tools, relevant industry certifications, and potential growth opportunities. This will help you connect academic concepts with real-world professional roles.

Task Description:

You will research and create a detailed report on at least 10 distinct cybersecurity domains. For each domain, you must address all seven of the points listed below. Your research must be thorough and accurately cited.

For each of the 10 domains you choose, you must include the following sections:

1. **Definition and Overview:** A clear, concise explanation of what the domain is and its primary purpose in the cybersecurity world.
2. **Key Responsibilities:** A list of the typical day-to-day tasks and duties for a professional in this role.
3. **Required Technical Skills:** The specific hard skills, knowledge areas, and technical proficiencies needed (e.g., programming languages, network protocols, operating system expertise).
4. **Required Soft Skills:** The non-technical skills crucial for success in the role (e.g., communication, problem-solving, attention to detail).
5. **Tools Used:** Common software, platforms, or utilities used by professionals in this domain.
6. **Certifications Related to the Domain:** At least two relevant industry certifications that validate expertise in this area.

7. Career Path & Growth Opportunities: A description of potential entry-level roles, advanced positions, and long-term career progression within this domain.

Top Domains Of Cyber Security :

1) **Penetration Testing / Ethical Hacking**
2) **Digital Forensics and Incident Response (DFIR)**
3) **Security Operations Center (SOC) Analyst**
4) **Malware Analysis / Reverse Engineering**
5) **Threat Intelligence**
6) **Cloud Security**
7) **Network Security**
8) **Application Security (AppSec)**
9) **Governance, Risk, and Compliance (GRC)**
10) **Red Team / Blue Team / Purple Team Operations**
11) **Threat Hunting**
12) **Identity and Access Management (IAM)**

Cyber Security Certifications :

1) **CompTIA:** Security+, CySA+, PenTest+
2) **EC-Council**: Certified Ethical Hacker (CEH), Computer Hacking Forensic Investigator (CHFI)
3) **Offensive Security:** OSCP, OSWP
4) **ISACA:** CISM, CISA, CRISC, CCOA
5) **(ISC)²:** CISSP, SSCP
6) **GIAC:** GSEC, GCFA, GCIH
7) **Cloud:** CCSP (Cloud), Certified Cloud Security Engineer (CCSE)

1. Penetration Testing / Ethical Hacking

Definition: Professionals paid to legally break into systems (networks, apps, devices) to find and fix security weaknesses before malicious hackers can exploit them .

Key Responsibilities: Conducting manual and automated security tests, reviewing code, writing detailed reports on vulnerabilities, and performing validation tests after fixes are applied .

Technical Skills: Deep understanding of network and web application security (e.g., SQLi, XSS), operating system exploitation, scripting (Python, Bash, PowerShell), cryptography, and cloud security .

Soft Skills: Critical thinking, problem-solving, creativity, strong communication (to explain technical issues to non-technical teams), and ethical judgment .

Tools Used: Burp Suite, Metasploit, Nmap, Wireshark, Nessus, Kali Linux .

Certifications: OSCP (Offensive Security), CEH (EC-Council), CompTIA PenTest+, GPEN (GIAC) .

Career Path: Can start as a network or SOC analyst, then progress to Junior Penetration Tester, Senior Penetration Tester, and Red Team Lead .

2. Digital Forensics and Incident Response (DFIR)

Definition: The practice of investigating and analyzing cyber incidents (like breaches) to understand the extent of the attack, contain it, and recover systems. It also involves collecting and preserving digital evidence for potential legal action .

Key Responsibilities: Performing forensics on networks, hosts, and memory; collecting and analyzing Indicators of Compromise (IOCs); maintaining a strict chain of custody for evidence; and possibly testifying in legal proceedings .

Technical Skills: Deep understanding of OS internals (Windows, macOS, Linux), experience with forensics tools, cloud services, and eDiscovery processes .

Soft Skills: Meticulous attention to detail, analytical thinking, strong written and verbal communication for reporting, and the ability to remain calm under pressure .

Tools Used: EnCase, FTK (Forensic Toolkit), Axiom, SIFT Workstation, Splunk, Volatility, Wireshark .

Certifications: EnCE (EnCase), GCFA (GIAC), CHFI (EC-Council), CCME (Magnet Forensics) .

Career Path: Digital Forensic Examiner, Incident Responder, Principal Investigator, DFIR Manager .

3. Security Operations Center (SOC) Analyst

Definition: A "digital consulting detective" who monitors an organization's security posture, analyzes log files for anomalies, and responds to potential threats in real-time .

Key Responsibilities: Monitoring security alerts, analyzing logs from various sources (firewalls, endpoints), identifying and containing threats, and communicating risks to business stakeholders .

Technical Skills: Proficiency with SIEM tools, understanding of attacker TTPs (Tactics, Techniques, Procedures) using frameworks like MITRE ATT&CK, log analysis, and incident containment procedures .

Soft Skills: Sharp analytical thinking, fast decision-making, clear communication, and a relentless drive to learn .

Tools Used: SIEMs (Splunk, ELK Stack), EDR platforms, firewalls, IDS/IPS .

Certifications: CompTIA Security+, CySA+, ISACA's CCOA (Certified Cybersecurity Operations Analyst), Cisco CyberOps Associate .

Career Path: SOC Analyst Level 1 -> Level 2 -> Level 3, SOC Manager, or specialize in Threat Hunting or Incident Response .

4. Malware Analysis

Definition: The process of dissecting malicious software to understand its functionality, origin, and potential impact. This can be done statically (examining the code without running it) or dynamically (running it in a safe, isolated environment) .

Key Responsibilities: Setting up secure lab environments, classifying malware types, analyzing behavioral characteristics, and documenting findings to improve detection and defense .

Technical Skills: Firm grasp of operating system internals (processes, memory, registry, APIs), experience with virtual machines, command-line administration, and familiarity with the Sysinternals suite .

Soft Skills: Patience, persistence, and strong problem-solving skills to unravel complex, obfuscated code.

Tools Used: Any.run, VirusTotal, IDA Pro, Ghidra, Sysinternals tools, Wireshark .

Certifications: GIAC Reverse Engineering Malware (GREM), SANS FOR610 related certifications.

Career Path: Malware Analyst, Reverse Engineer, Threat Researcher.

5. Threat Intelligence

Definition: The process of collecting and analyzing information about emerging threats, attack patterns, and malicious actors to help an organization proactively defend itself .

Key Responsibilities: Gathering data on threats, analyzing attack patterns (TTPs) using frameworks like MITRE ATT&CK, and disseminating actionable intelligence to security teams .

Technical Skills: Data analysis, understanding of cyber threat landscapes, proficiency with threat intelligence platforms, and knowledge of various attack vectors.

Soft Skills: Analytical and critical thinking, strong writing skills for creating reports, and the ability to connect disparate data points to form a coherent picture.

Tools Used: Threat Intelligence Platforms (TIPs), SIEM, OSINT tools (Maltego, Spiderfoot) .

Certifications: GIAC Cyber Threat Intelligence (GCTI), Certified Threat Intelligence Analyst (C|TIA).

Career Path: Threat Intelligence Analyst, Threat Researcher, Intelligence Manager.

6. Cloud Security

Definition: The practice of protecting cloud-based systems, data, and infrastructure. This includes configuring and managing security for platforms like AWS, Azure, and Google Cloud .

Key Responsibilities: Securing cloud infrastructure, managing identity and access management (IAM), implementing data protection, and conducting cloud penetration tests and forensic investigations .

Technical Skills: Understanding of cloud architecture (AWS, Azure, GCP), IAM, data security, cloud incident response, and business continuity planning in the cloud .

Soft Skills: Adaptability (as cloud tech changes fast), collaboration with development and operations teams, and a deep understanding of compliance.

Tools Used: AWS Security Hub, Azure Security Center, Prisma Cloud, Cloud workload protection platforms (CWPP) .

Certifications: Certified Cloud Security Engineer (CCSE), CCSP ((ISC)²), AWS Certified Security – Specialty .

Career Path: Cloud Security Engineer, Cloud Security Architect, Cloud Compliance Analyst.

7. Network Security

Definition: The practice of designing, implementing, and managing security measures to protect the integrity, confidentiality, and accessibility of computer networks and devices .

Key Responsibilities: Configuring and managing next-generation firewalls, implementing VPNs for secure remote access, setting up IDS/IPS, and designing IAM mechanisms to protect network assets .

Technical Skills: Deep knowledge of networking protocols (TCP/IP), configuring routers and switches, firewall rule management, and network access control .

Soft Skills: Systematic thinking, troubleshooting complex issues, and meticulous attention to detail to avoid misconfigurations.

Tools Used: Firewalls (Palo Alto, Fortinet), Wireshark, Nmap, VPN gateways .

Certifications: CompTIA Security+, PCNSE (Palo Alto Networks), CCNA (Cisco) .

Career Path: Network Administrator, Network Security Engineer, Firewall Administrator, Network Security Architect .

8. Application Security (AppSec)

Definition: The practice of incorporating security throughout the software development lifecycle (SDLC) to prevent vulnerabilities in applications. This is often summarized as "security by design" .

Key Responsibilities: Performing defensive coding, conducting vulnerability assessments and penetration tests on applications, using OWASP principles to guide development, and implementing continuous monitoring for applications in production .

Technical Skills: Secure coding practices, knowledge of OWASP Top 10, experience with vulnerability scanning tools, and familiarity with DevOps/CI/CD pipelines .

Soft Skills: Ability to collaborate with developers, patience in explaining security flaws, and a focus on enabling development rather than blocking it.

Tools Used: Burp Suite, OWASP ZAP, Static/Dynamic Application Security Testing tools (SAST/DAST) .

Certifications: Certified Application Security Engineer (CASE), GIAC Web Application Penetration Tester (GWAPT).

Career Path: Application Security Engineer, DevSecOps Engineer, Application Security Architect.

9. Governance, Risk, and Compliance (GRC)

Definition: The "business backbone" of cybersecurity. GRC professionals ensure an organization's security program has a solid foundation through policies (Governance), identification and mitigation of business risks (Risk), and adherence to laws and regulations (Compliance) .

Key Responsibilities: Creating and maintaining security policies, conducting risk assessments, ensuring compliance with standards like GDPR or HIPAA, and acting as a bridge between technical teams and business leadership .

Required Skills: Business acumen, understanding of frameworks (ISO 27001, NIST CSF), policy development, and stakeholder management. Minimal deep technical skills are required .

Soft Skills: Excellent communication, strong writing skills, the ability to translate technical risks into business language, and critical thinking .

Tools Used: Governance, risk management, and compliance (GRC) platforms, policy management tools, risk registers.

Certifications: CISA, CISM, CRISC (all from ISACA), CISSP .

Career Path: GRC Analyst, Compliance Officer, Risk Manager, Chief Information Security Officer (CISO) .

10. Threat Hunting

Definition: The proactive and iterative process of searching through networks, endpoints, and datasets to detect malicious, suspicious, or risky activities that have evaded existing security tools .

Key Responsibilities: Developing hypotheses about potential threats, using threat intelligence to guide hunts, using data analytics to find anomalies, and collaborating with SOC and IR teams .

Technical Skills: Proficiency with SIEM and EDR tools, scripting (Python, PowerShell), data analysis, and deep knowledge of attacker TTPs .

Soft Skills: Curiosity, creativity, persistence, and an "attacker mindset" to think about creative threats and attack vectors .

Tools Used: SIEM platforms, EDR tools, network traffic analyzers (Wireshark), and big data platforms like Elastic Stack .

Certifications: GIAC Certified Threat Hunter (GCTH), OSCP, and other advanced security certifications .

Career Path: Threat Hunting Analyst, Senior Threat Hunter, Threat Hunting Lead.

11. Red Team / Blue Team Operations

Definition: A team-based approach to security. The Red Team acts as adversaries to simulate real-world attacks and test defenses. The Blue Team defends against the Red Team's attacks, monitoring, detecting, and responding. Purple Team involves close collaboration between both to maximize the effectiveness of the exercise .

Key Responsibilities (Red Team): Simulating advanced attacks, using TTPs of real threat actors, and attempting to breach physical, digital, and human defenses without being detected .

Key Responsibilities (Blue Team): Continuously monitoring for the Red Team's activities, improving detection capabilities, and refining incident response processes based on the exercise.

Technical Skills: Red teamers need advanced exploitation and evasion skills; blue teamers need advanced detection, analysis, and defense skills.

Soft Skills: For red teams: creativity and stealth. For blue teams: vigilance and resilience. For both: excellent communication and collaboration.

Tools Used: Red teams use tools like Cobalt Strike, Metasploit. Blue teams use SIEM, EDR, and other defensive tools.

Certifications: OSCP, CRTP (Red), GCFA, GCIA (Blue).

Career Path: Red Team Operator, Blue Team Lead, Purple Team Engineer, CISO.

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